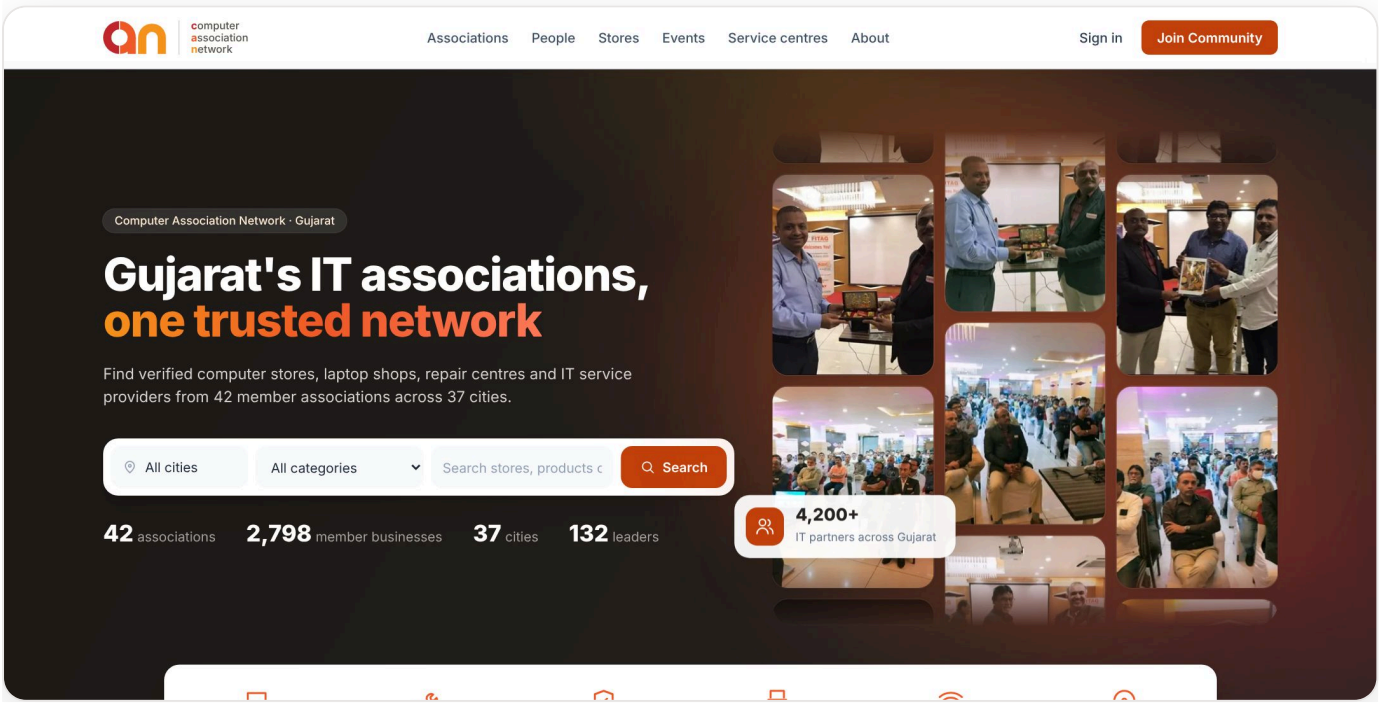
computer
association
network

Platform demo walkthrough & future scope

Computer Association Network (CAN) — one platform for Gujarat's IT trade associations and their member businesses, built for the member associations of FITAG.

42	2,798	37	132
member associations	member businesses listed	cities in 6 zones	leaders with profiles

Version 1 demo · November 2026



OVERVIEW

What CAN is

CAN connects every IT trade association in Gujarat and their member shops on one platform. Customers find trusted local computer stores; each member business gets its own branded website and a complete shop-management system; the community team runs the whole network from one admin panel. Each store's customers and financial data stay private to that store.

1 · Community website

Associations by zone, leaders, store directory, events, brand service centres, membership sign-up.

2 · Store website

Each member's own branded site on its own domain: products, services, online booking, repair tracking.

3 · Store ERP & CRM

POS and GST billing, serial-level inventory, purchases, repair job cards, bookings, customers, reports.

4 · Master admin

Approve members, manage cities, areas and plans, suspend or verify stores, network dashboard, audit log.

In this document

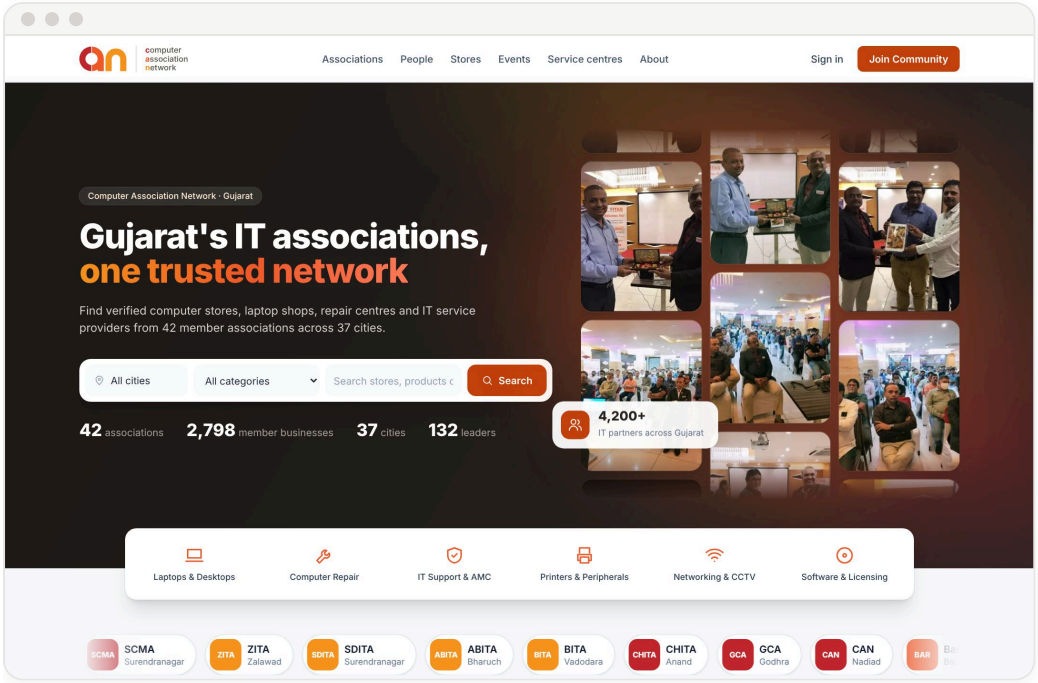
1. Community website — the network's public face
2. Joining the network — application to approved workspace
3. Store ERP & CRM — running the shop
4. Store website — the customer's view
5. Master admin — running the network
6. 15-minute demo script
7. Under the hood — security, data, deployment
8. What is not built yet
9. Future scope — phased roadmap

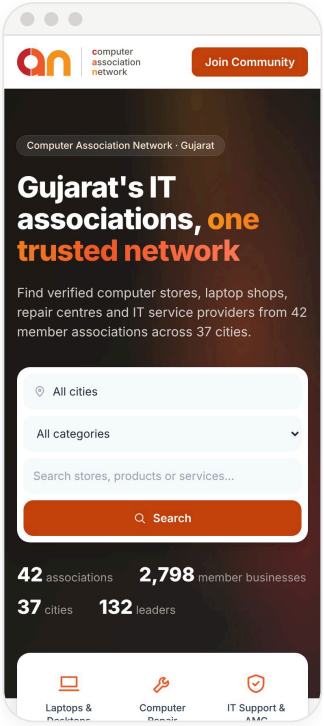
Demo access. Community site: `https://mcvc.201-18-199-217.sslip.io` (test address until a domain is connected). Demo store owner: `owner@techworld.local` with the demo password set on the server (`SEED_PASSWORD`). Admin: the email and password configured at deployment (`ADMIN_EMAIL`).

1 · COMMUNITY WEBSITE

Home page

The first thing a visitor sees: who the network is, how big it is, and a search for any store, product or service.





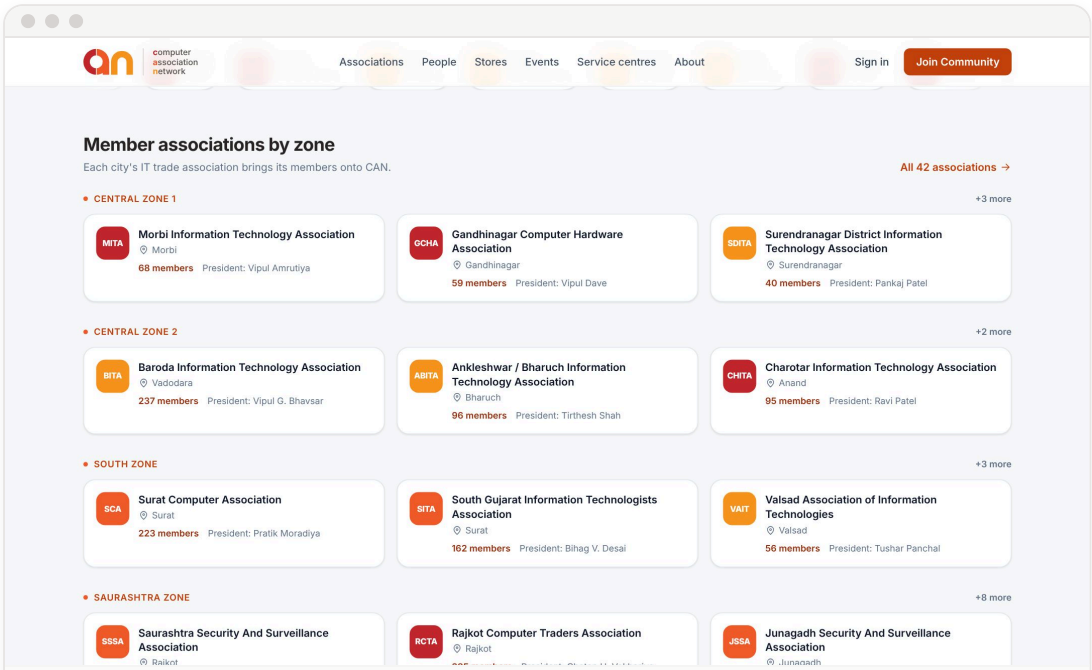
- **Animated photo wall** of real FITAG events and members; counters animate on load.
- **Search** by city, category and keyword; six category shortcuts below.
- **Association ribbon** scrolls every association code — click any to open its page.
- Fully responsive; animation switches off for visitors who prefer reduced motion.

> **Try it:** open the site, type “laptop repair” and choose Nadiad.

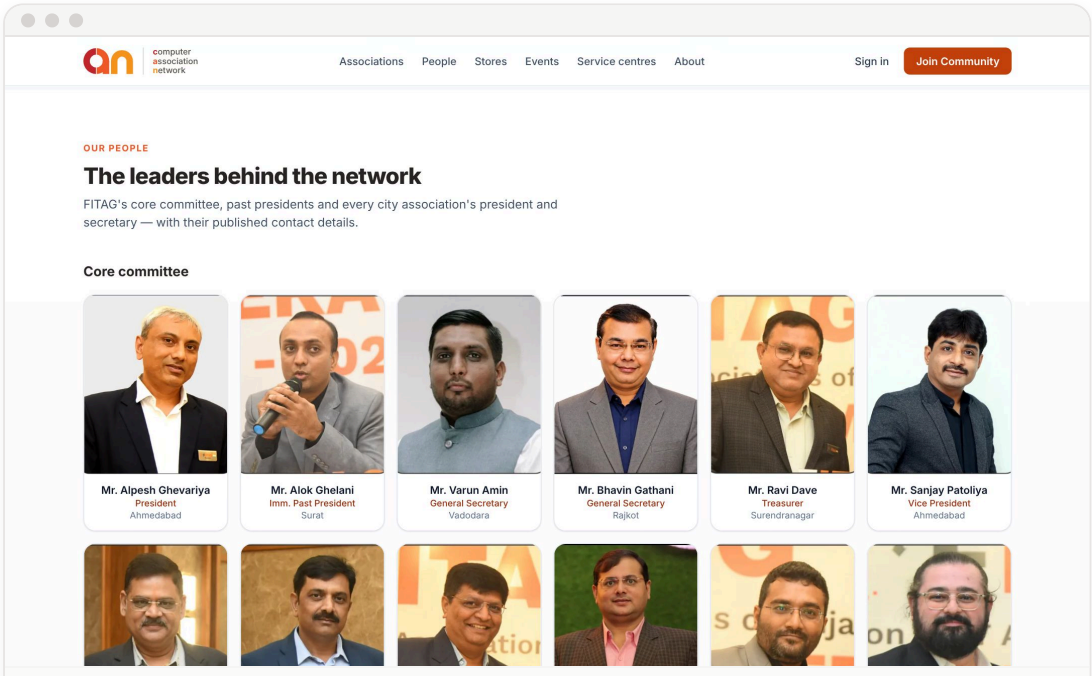
1 · COMMUNITY WEBSITE

Associations by zone and the people behind them

The home page puts the associations and their leadership up front, taken from the parent body's published information.



Associations grouped into Central 1 & 2, South, Saurashtra, North and Kutch zones



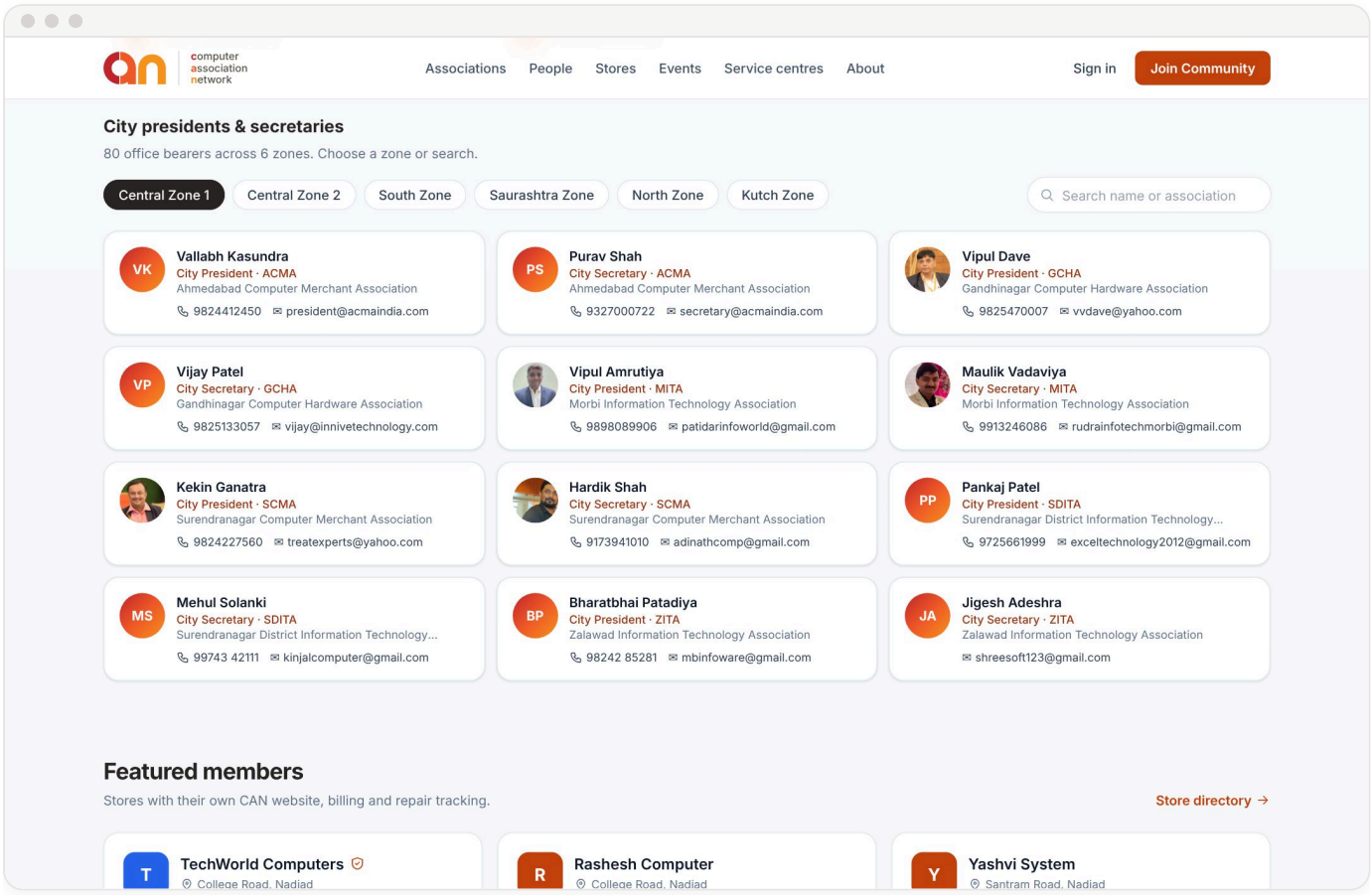
Core committee with photos, roles and cities

- 42 associations with member counts and their president.
- FITAG core committee (25), past presidents with years (8) and advisory committee.
- Photos are used only where the source names the person; everyone else shows initials.

1 · COMMUNITY WEBSITE

City presidents & secretaries

Every city association's president and secretary in one place, by zone, with the contact details the association publishes.

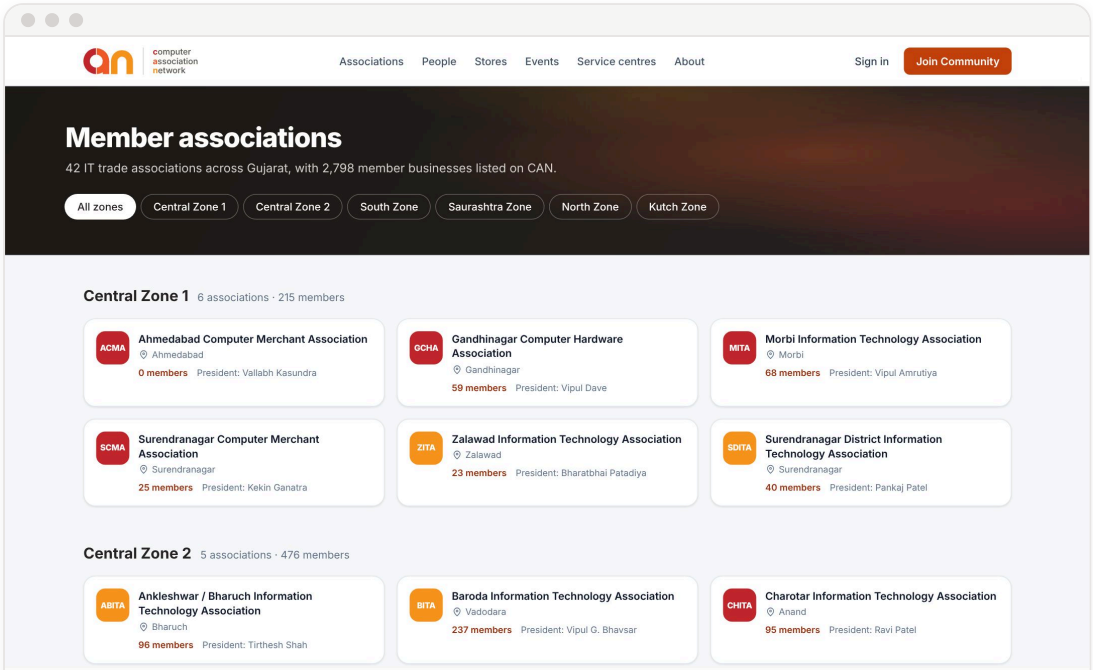


- Zone tabs and instant search by name, association or code.
- Tap-to-call phone and email links; each card links to the association page.
- 80 office bearers across 6 zones.

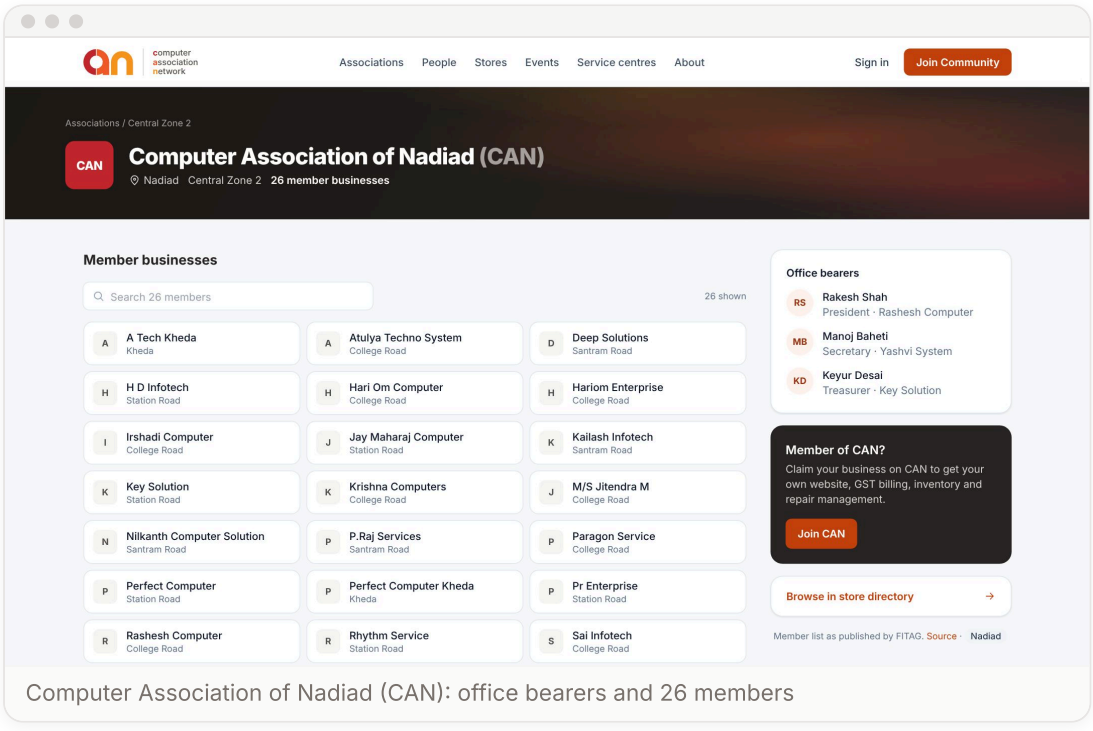
1 · COMMUNITY WEBSITE

Association pages

Each association gets its own page — the digital home for that city's IT trade.



All associations, filterable by zone



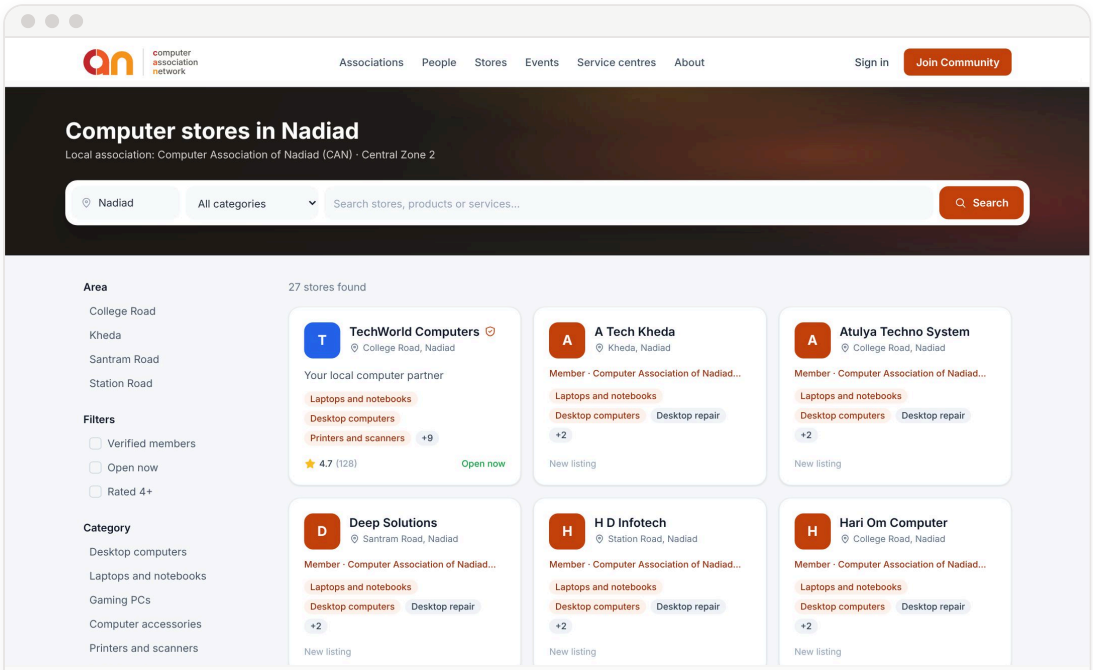
Computer Association of Nadiad (CAN): office bearers and 26 members

- Office bearers, member businesses with search, link into the store directory.
- “Join CAN” pre-selects the association in the membership form.

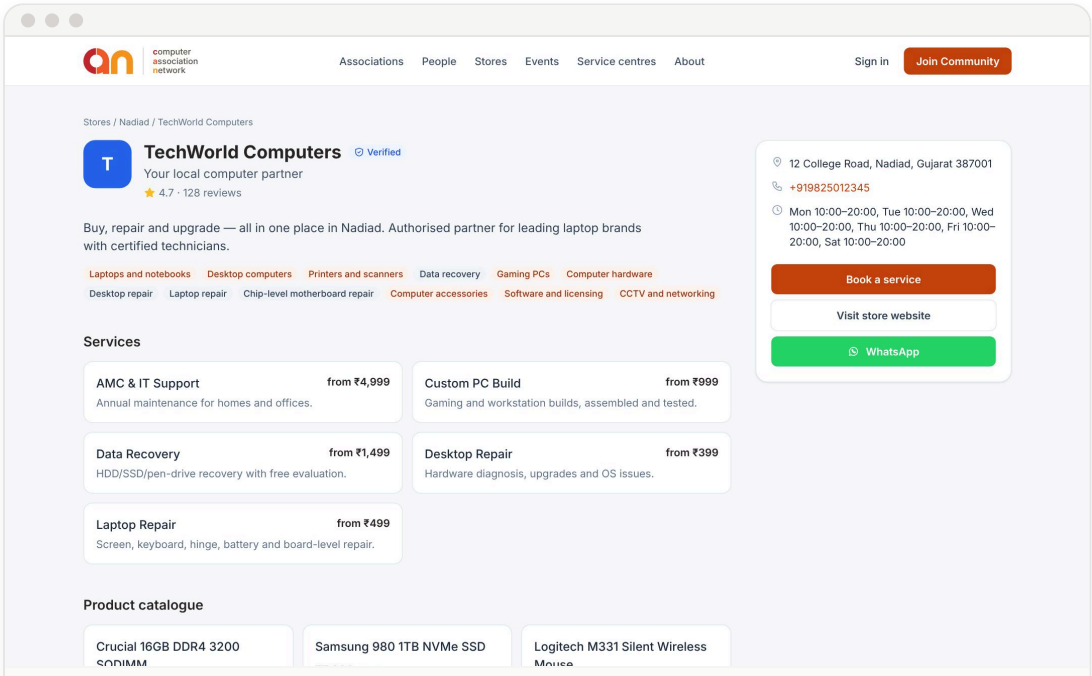
1 · COMMUNITY WEBSITE

Store directory and listings

Customers search across all member businesses. Members who have activated CAN show a full profile; others appear as name-only listings.



Directory: filters for area, category, verified, open now, rating, product availability



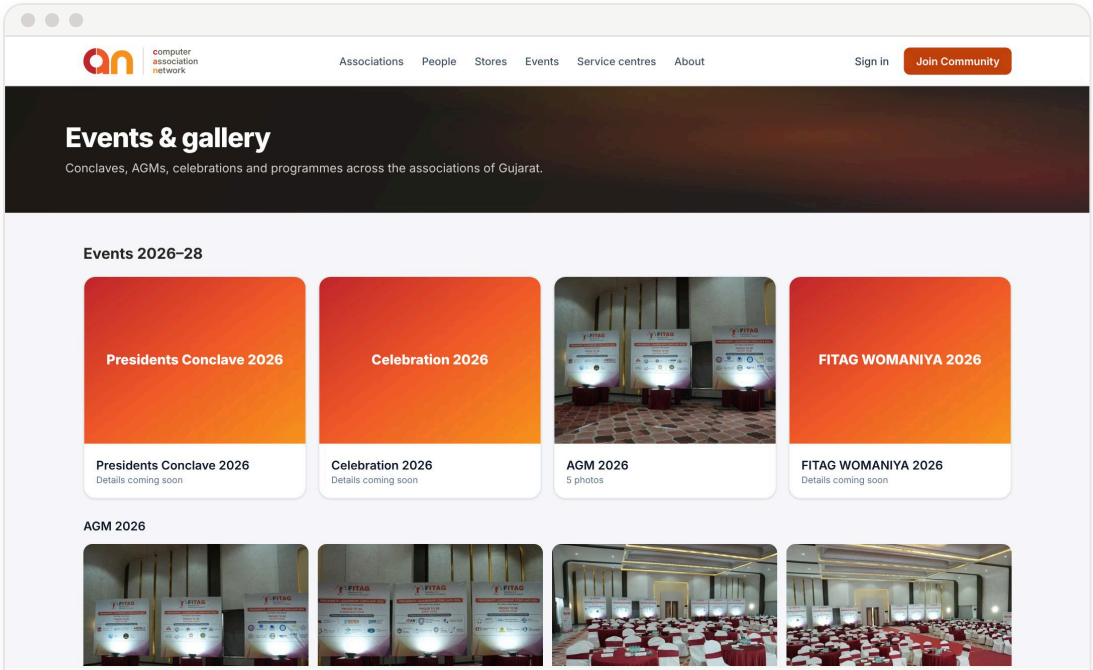
Full member profile: services, products with stock, hours, booking and WhatsApp

- 2,798 member firms are listed by name under their association — no personal phone numbers or GST numbers are copied.
- Each name-only listing has a “Claim it on CAN” link so the owner can complete it.
- Product search finds stores that have an item in stock (e.g. “RTX 4060”).

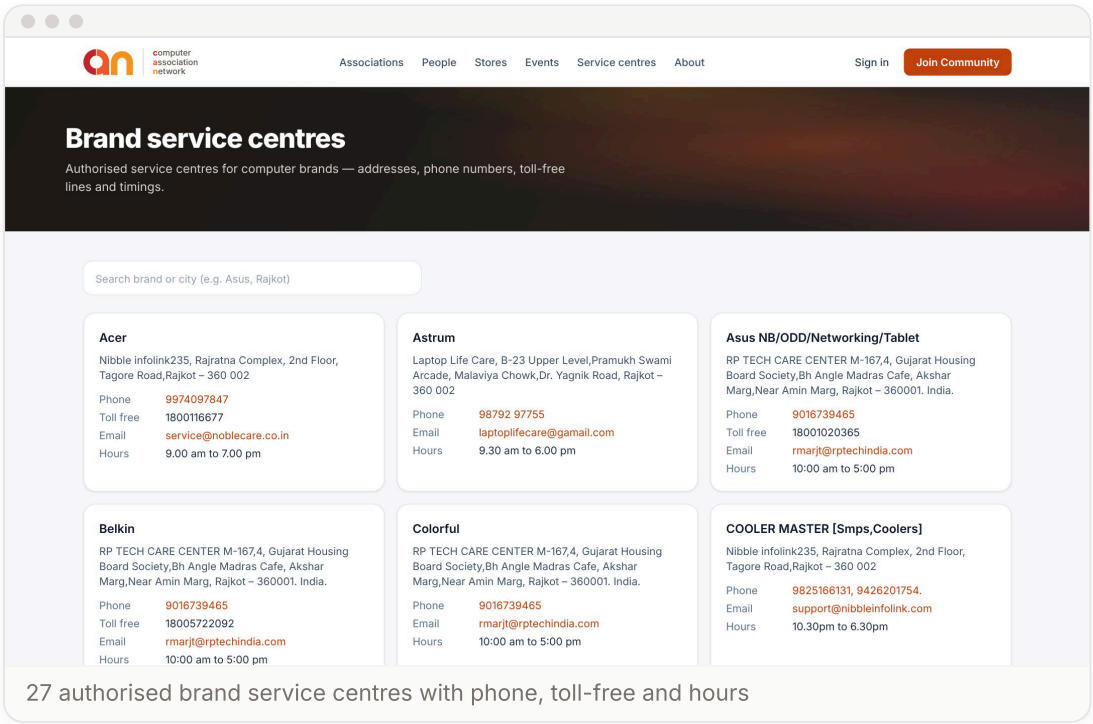
1 · COMMUNITY WEBSITE

Events, gallery and brand service centres

Useful content for members and customers, brought over from the parent body's site.



Events 2026–28 and association programmes

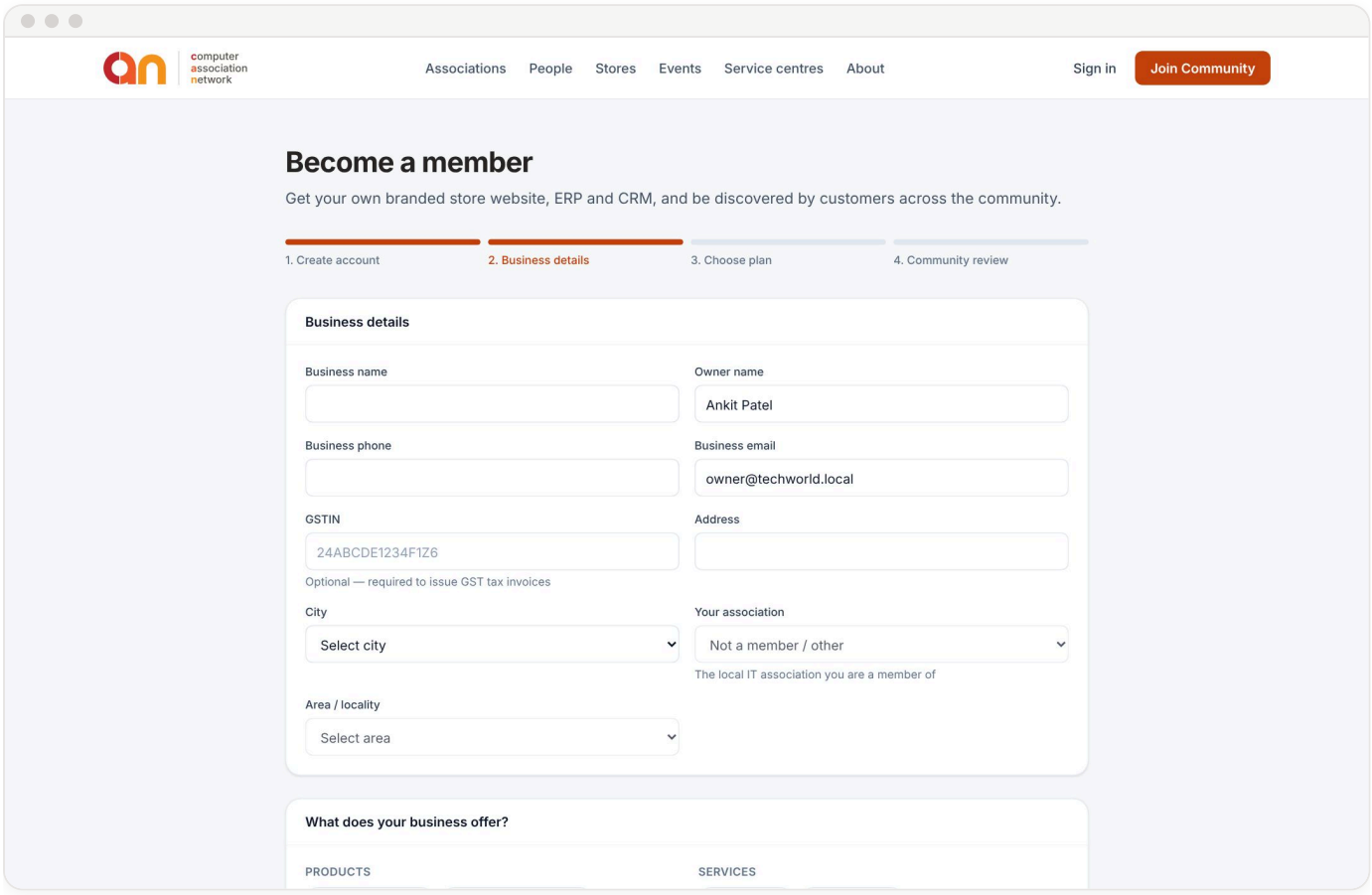


27 authorised brand service centres with phone, toll-free and hours

- AGM 2026 photos and the gallery of association programmes.
- Service centres searchable by brand or city (Asus, Dell, Intel, Kingston…).
- About page: FITAG profile, Five Point Agenda, objectives, full team, useful links and contacts.

Membership application

A shop owner joins in a few minutes. The community team reviews the application; approval creates the store's complete workspace automatically.

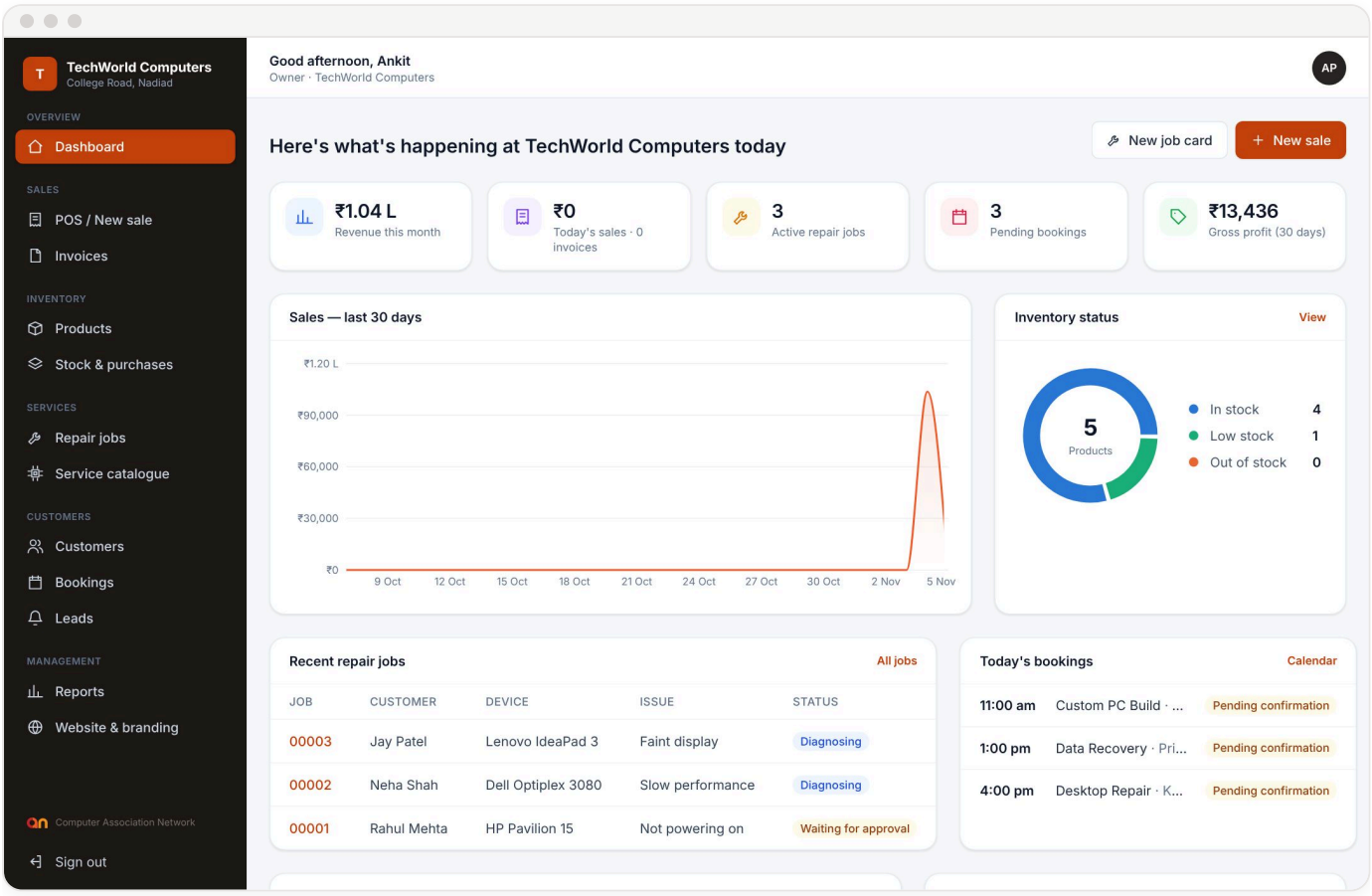


- Steps: create account → business details, city, area and **their association** → what they sell and service → choose a plan.
- GSTIN is checked (format and check digit). Indian mobile numbers are validated.
- On approval the system creates: the store, its staff roles (Owner, Manager, Cashier, Technician, Front Desk), settings, a 14-day trial subscription and a stock location.
- Membership approval and subscription/billing are separate, as the spec requires.

> **Try it:** register a test business, then approve it from the admin panel (section 5).

Store dashboard

The owner's morning view: today's sales, monthly revenue, profit, open repair jobs, bookings and stock that needs attention.

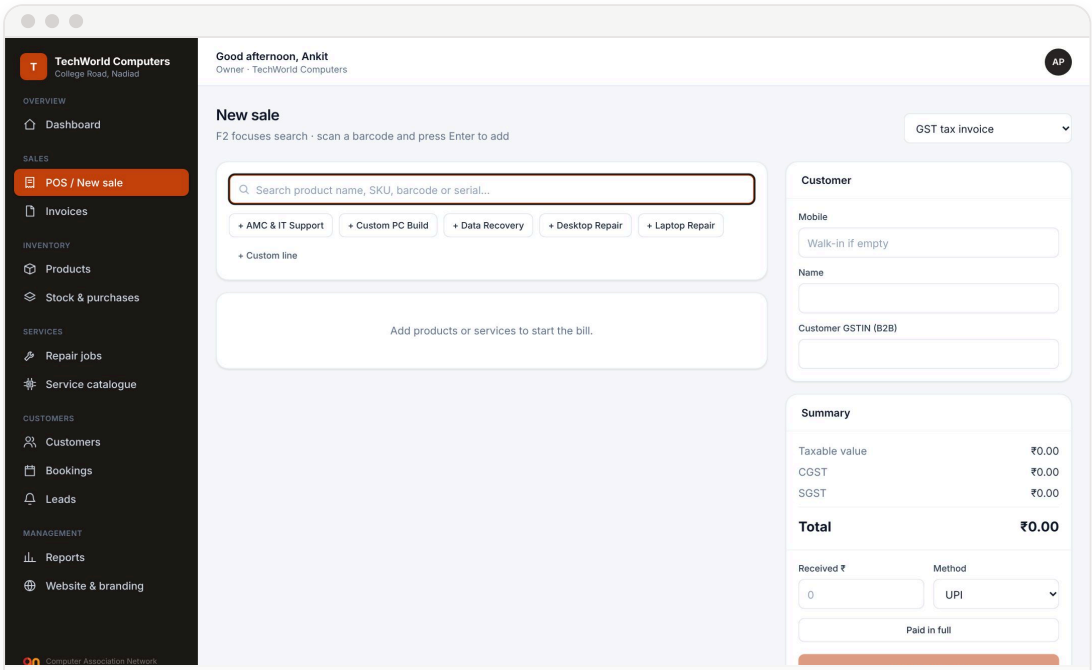


- Sales chart for the last 30 days, inventory status, recent repair jobs and today's bookings.
- Receivables, new leads, warranties expiring in 30 days and low-stock items.
- Each staff member sees only what their role allows.

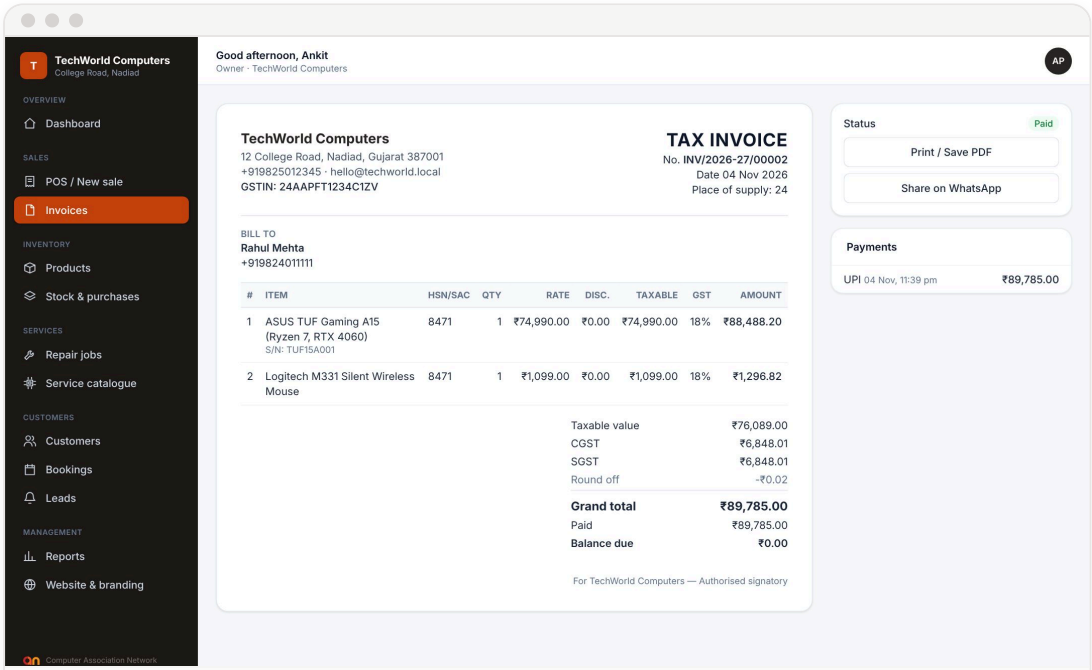
3 · STORE ERP & CRM

POS and GST invoices

Built for speed at the counter: barcode scanner support (scan + Enter), keyboard shortcut F2, one-tap “paid in full”.



New sale: scan or search, pick serial numbers, take payment



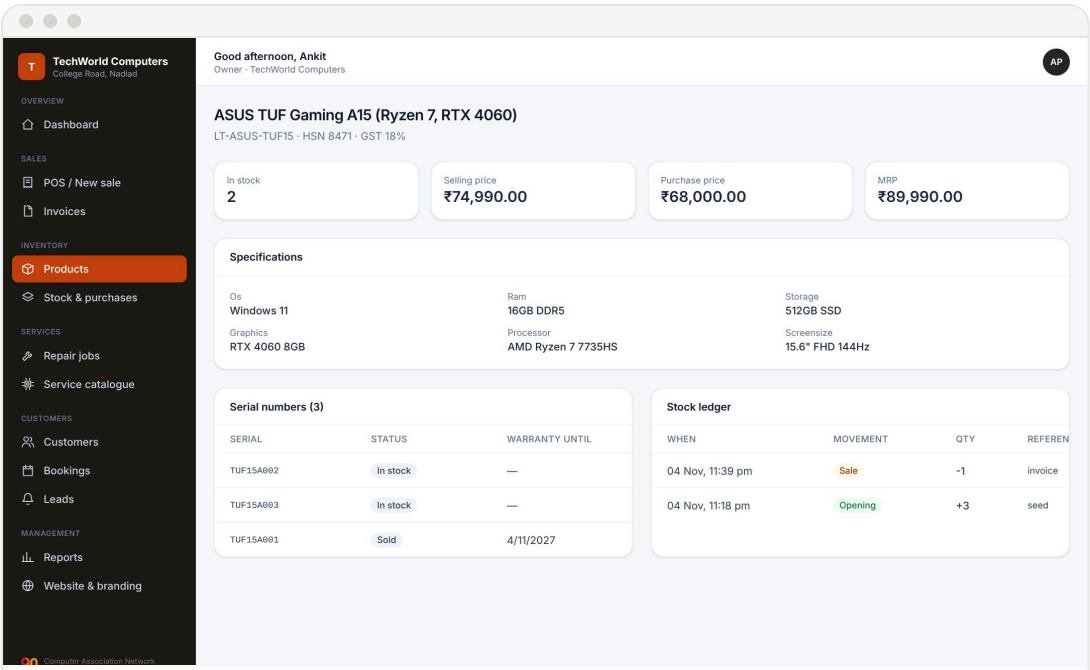
Printable GST tax invoice with CGST/SGST (or IGST) and payments

- Tax invoice, bill of supply, quotation and proforma; numbering per financial year (INV/2026-27/00001).
- CGST+SGST within Gujarat, IGST for other states (from the customer's GSTIN); tax-inclusive or exclusive prices; rupee round-off.
- Partial and multiple payments; cancelling restores stock and serial numbers. Print / save PDF and share on WhatsApp.
- GST rates allowed per store are configurable — and must be confirmed by a tax adviser.

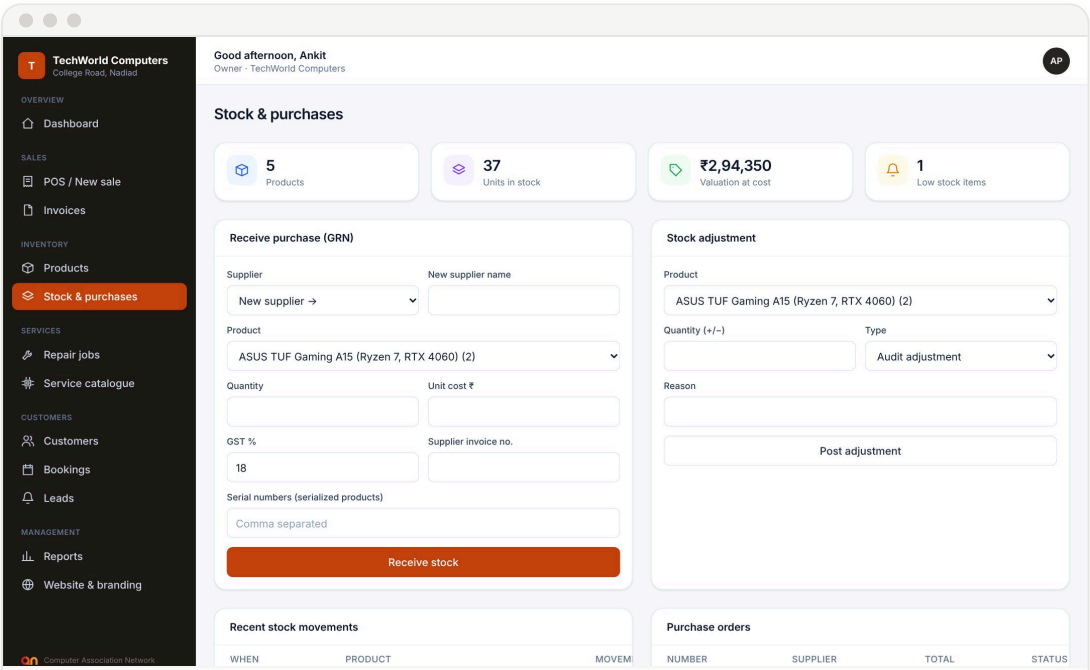
3 · STORE ERP & CRM

Products, serial numbers and stock

Computer retail needs serial-level tracking. Every stock change is recorded in a ledger that cannot be edited.



Product with computer specs, serial numbers and stock ledger



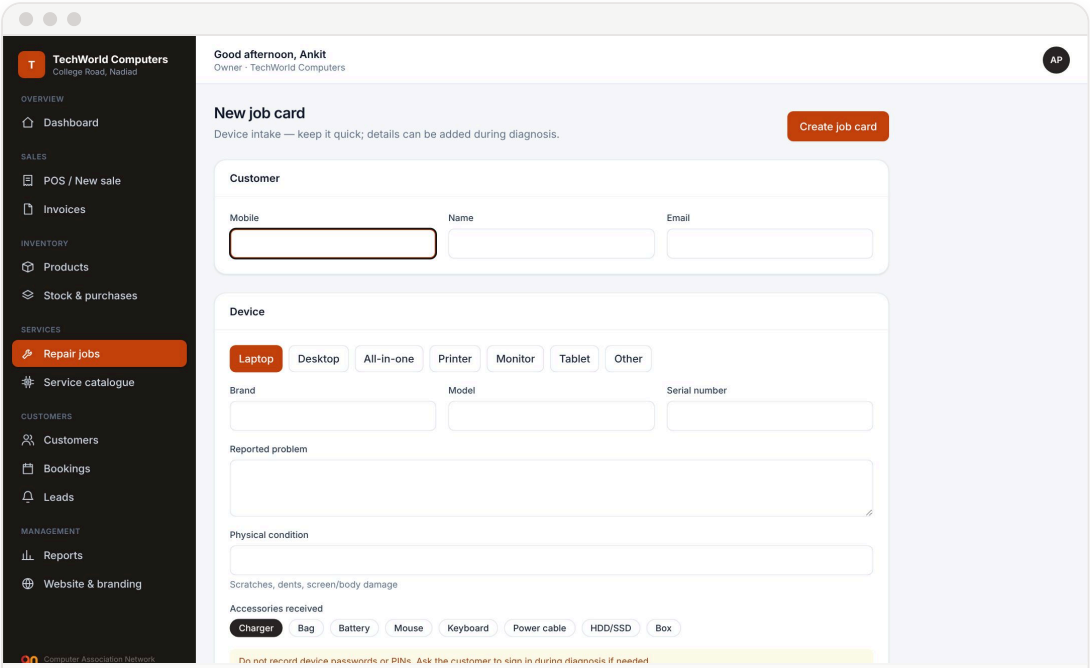
Purchases (GRN), stock adjustments, movements and valuation

- Processor, RAM, storage, GPU, screen, OS; MRP, purchase and selling price; HSN and GST; warranty months.
- Each laptop's serial/IMEI is tracked: in stock → sold (with warranty dates) → returned.
- Purchase receiving updates stock and cost price; damaged/audit adjustments need a reason.

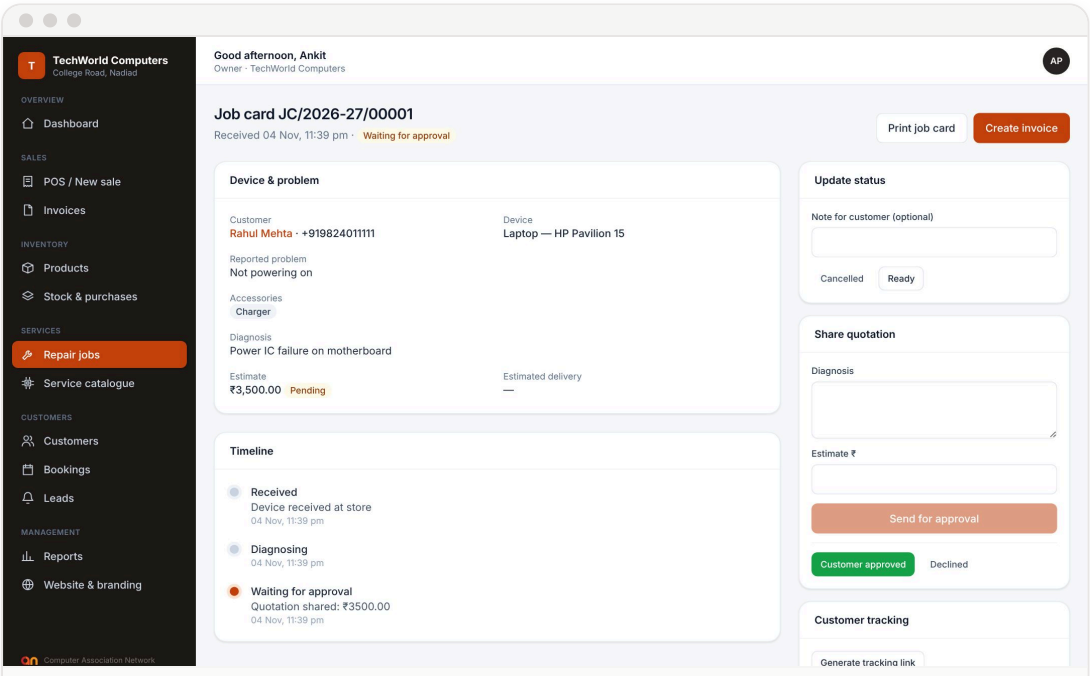
3 · STORE ERP & CRM

Repair job cards

The repair workflow from the spec, enforced step by step: Received → Diagnosing → Waiting for approval → In progress → Quality check → Ready → Delivered.



Fast device intake: customer, device, problem, accessories



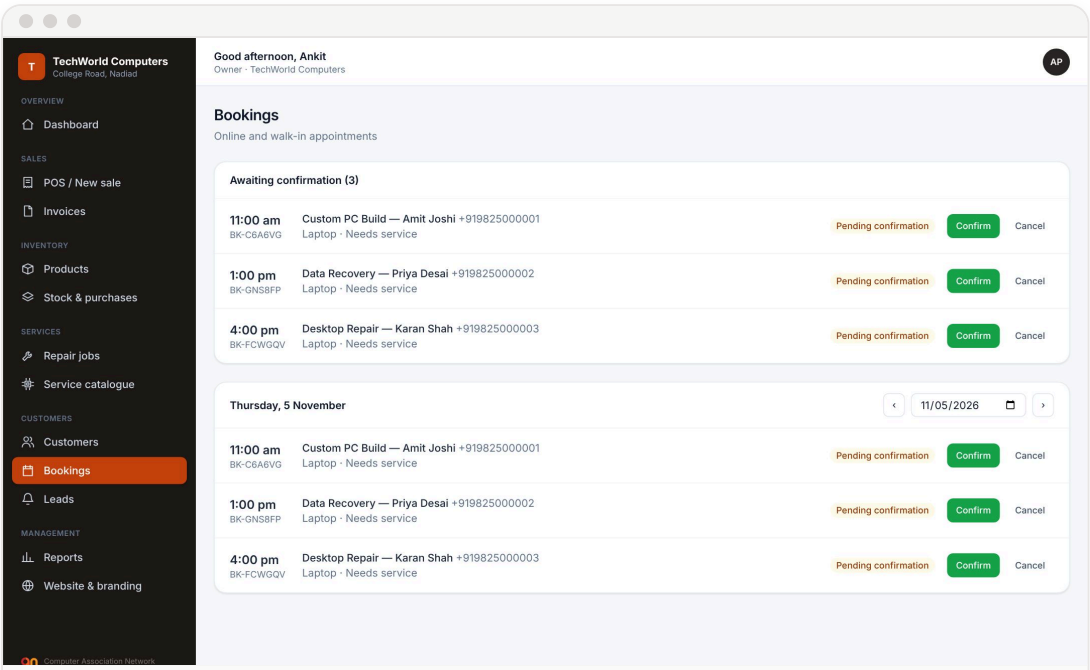
Job detail: timeline, quotation, parts, customer tracking link

- Job card numbers per financial year (JC/2026-27/00001). Device passwords are deliberately not stored.
- Quotation sent to the customer, who approves or declines online from the tracking link.
- Parts used are deducted from stock. Delivery is blocked until the invoice is paid (or marked no-charge).
- WhatsApp/SMS notifications are queued at each important step.

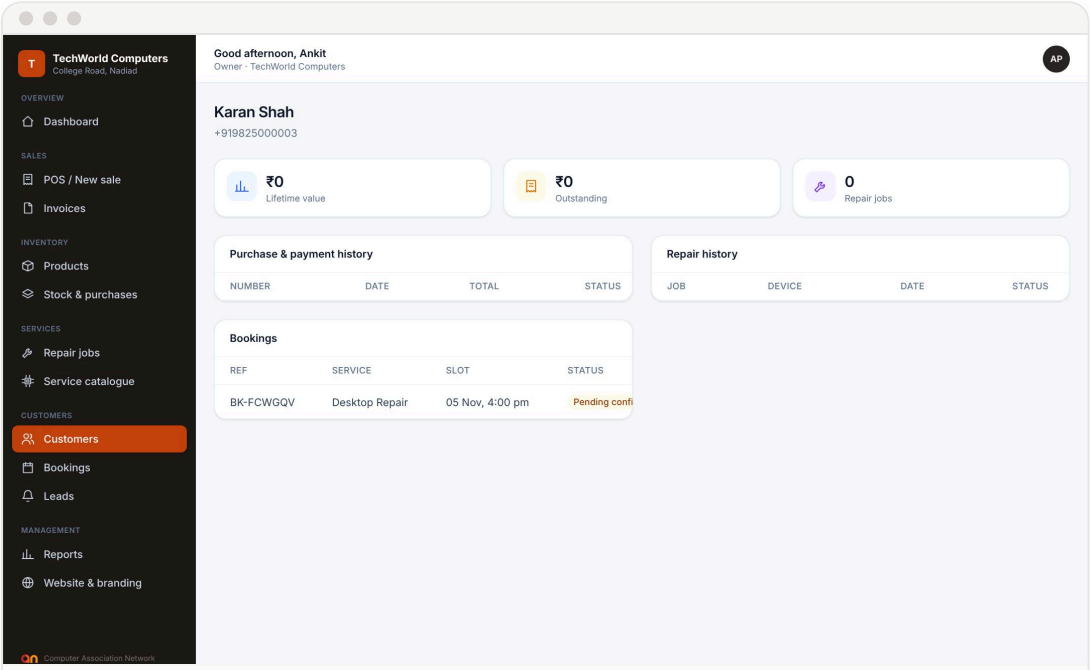
3 · STORE ERP & CRM

Bookings and customers

Bookings from the store website arrive here. Customers are searchable by name, phone, email, invoice number or device serial number.



Bookings by day; confirm, cancel or turn into a job card

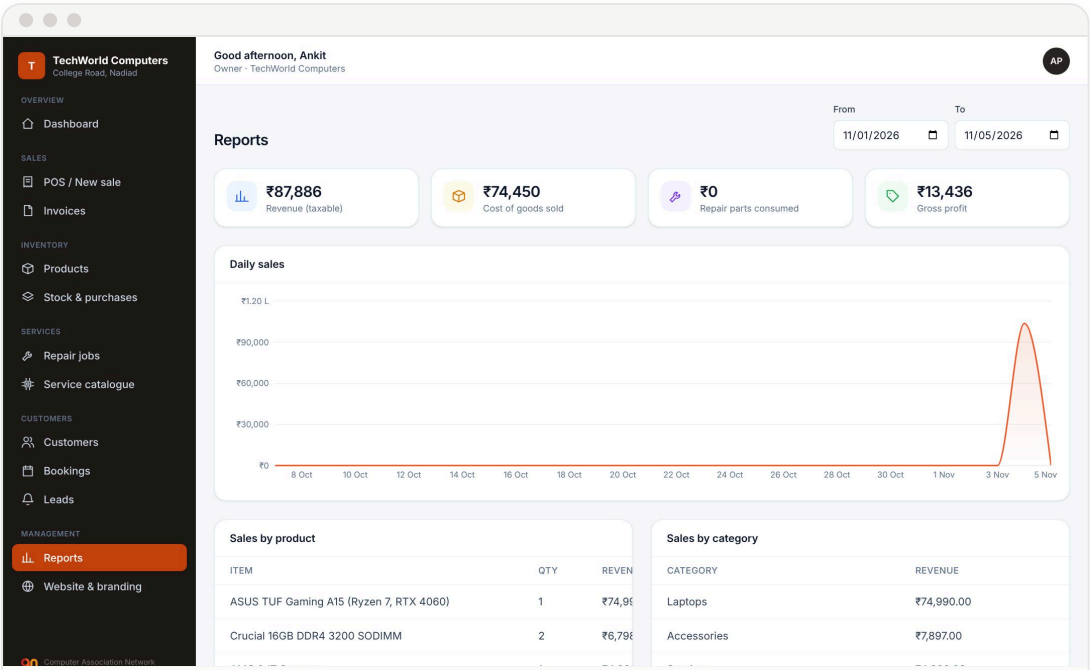


Customer profile: lifetime value, outstanding, purchase, repair and booking history

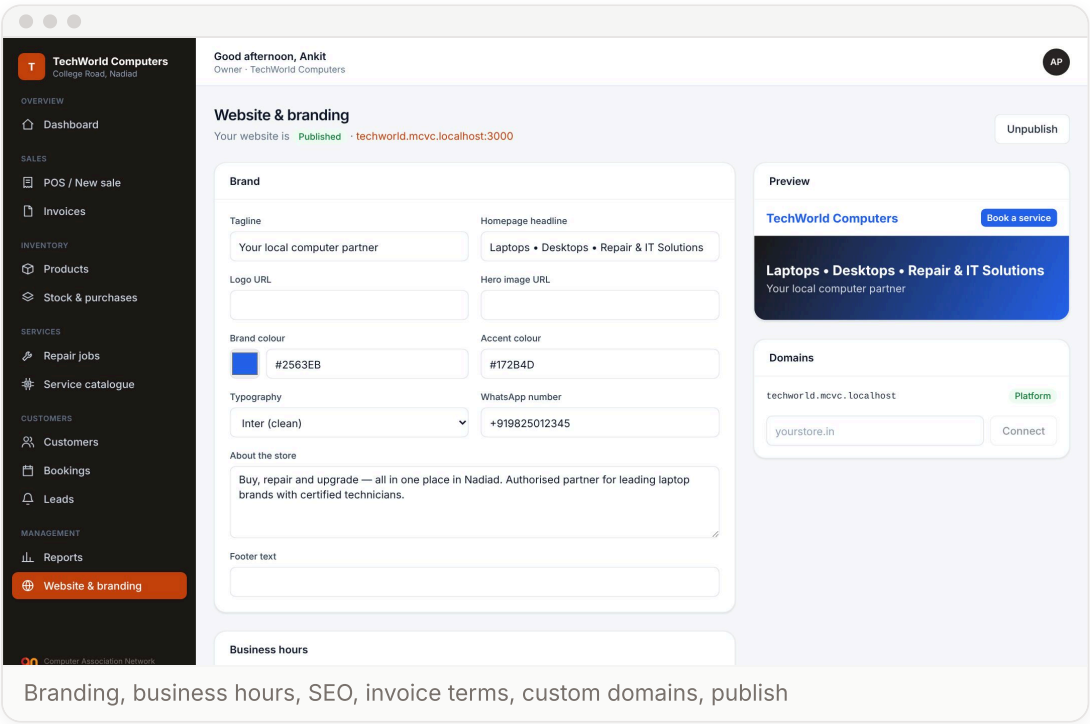
- Slot capacity per store: two customers can never take the last slot at the same time (tested with simultaneous requests).
- Leads from website enquiries with follow-up status.
- Marketing consent recorded per customer.

Reports and website settings

Owners see their numbers and control their own website without technical help.



Sales by product and category, gross profit, GST summary by rate and HSN



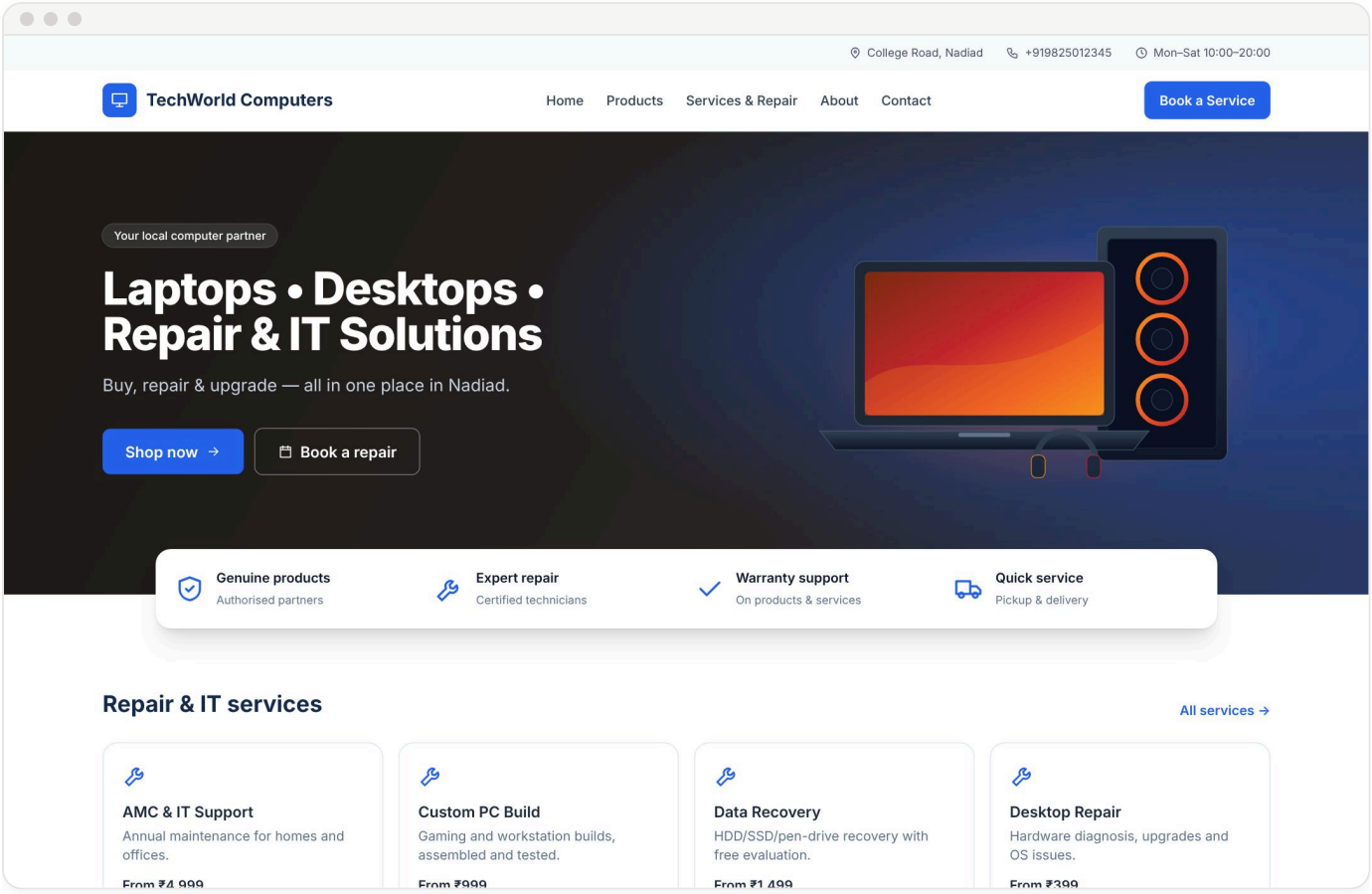
Branding, business hours, SEO, invoice terms, custom domains, publish

- Reports: daily sales, gross profit, repair parts consumed, GST summary (for review with a tax adviser).
- Branding: logo, colours, fonts, hero, WhatsApp, hours; live preview.
- Custom domain: add domain → DNS instructions → verify ownership → becomes the primary address with automatic SSL.

4 · STORE WEBSITE

The member's own website

Every member store gets a professional website in its own colours, on its own domain (e.g. techworld.in) or a free CAN sub-address.

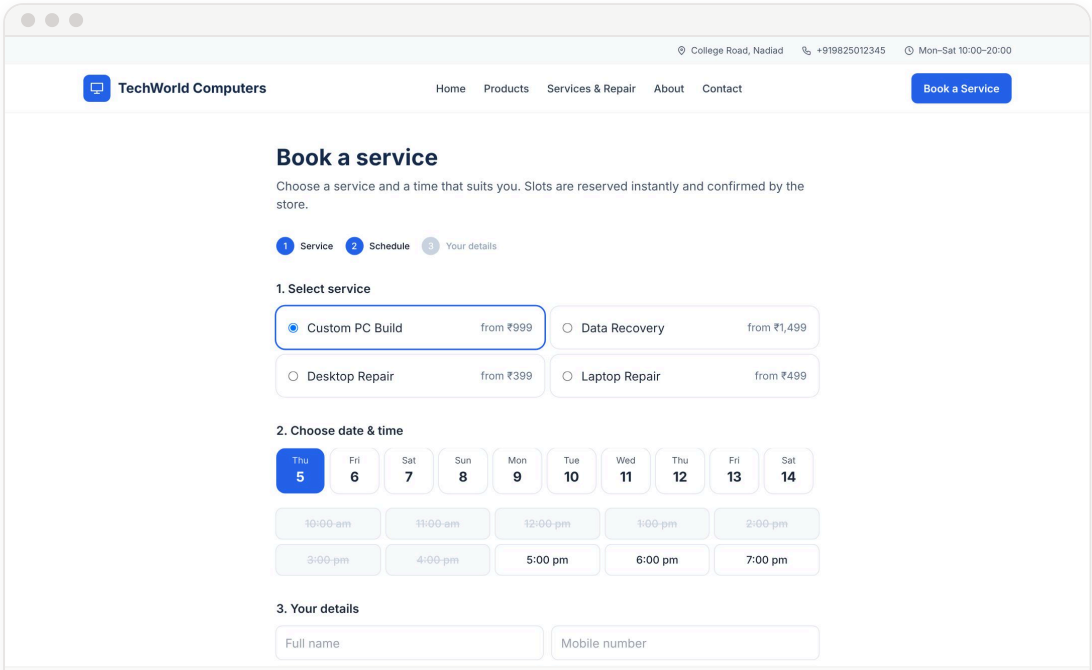


- Home, products, product details with specs, services, about, contact, WhatsApp.
- Product enquiries become leads in the store's CRM.
- One store's changes never affect another store's site.

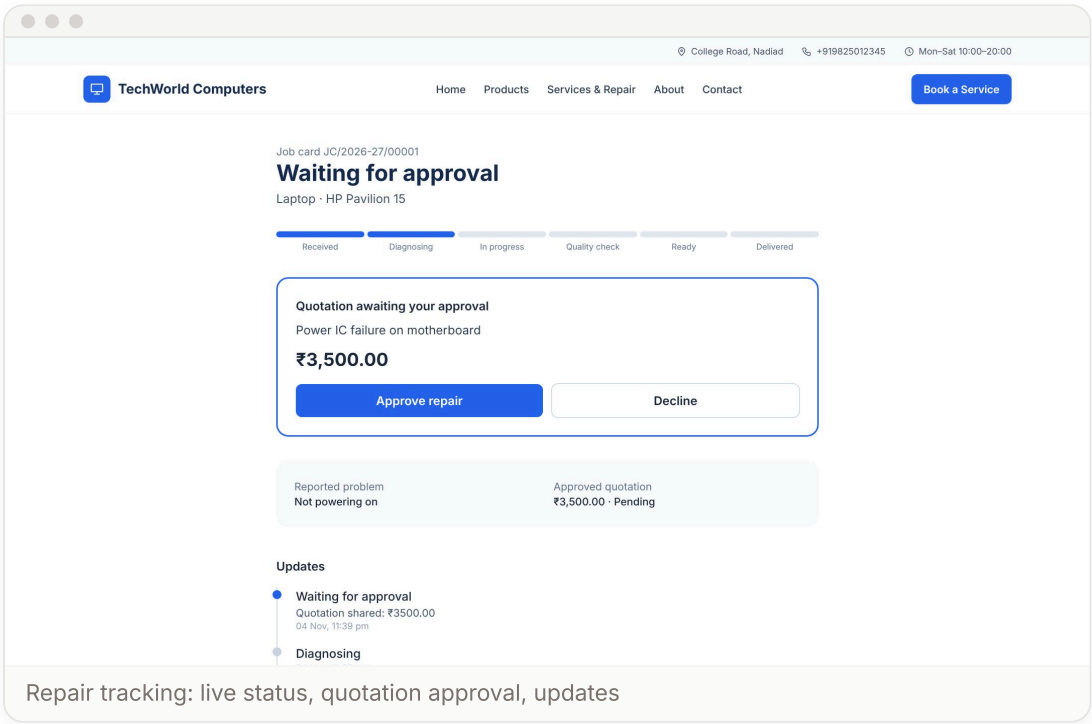
4 · STORE WEBSITE

Online booking and repair tracking

Customers book and follow their repair without calling the shop.



Book a service: choose service, date and an available time



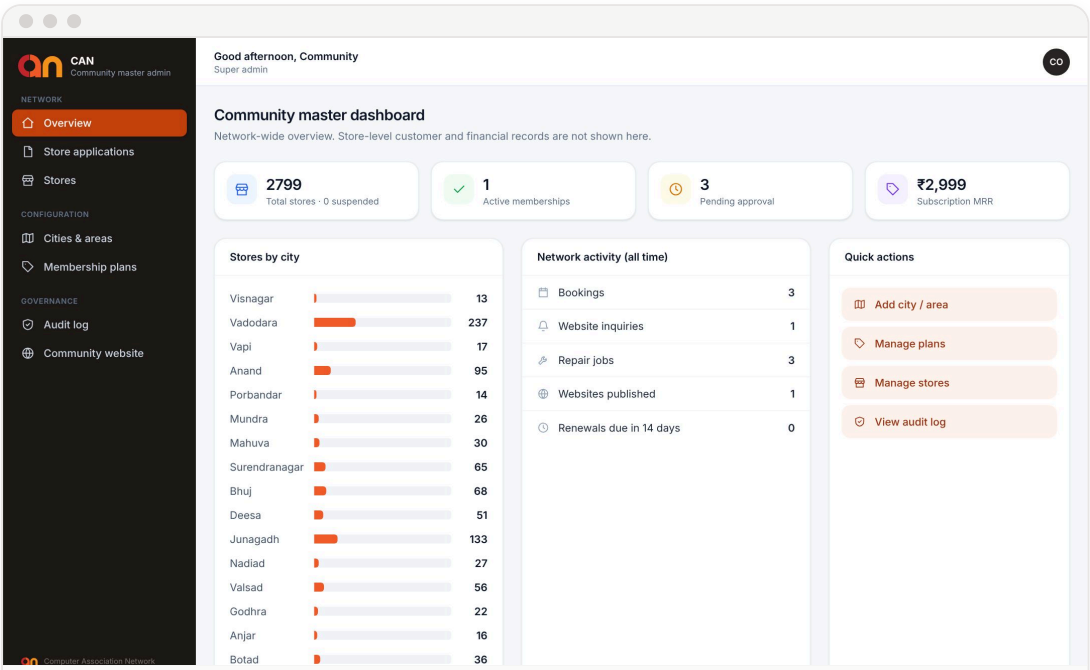
Repair tracking: live status, quotation approval, updates

- Only real, free slots inside business hours are offered.
- The tracking link is private and unguessable; it shows only customer-facing updates.
- Customers can approve or decline the repair quotation in one tap.

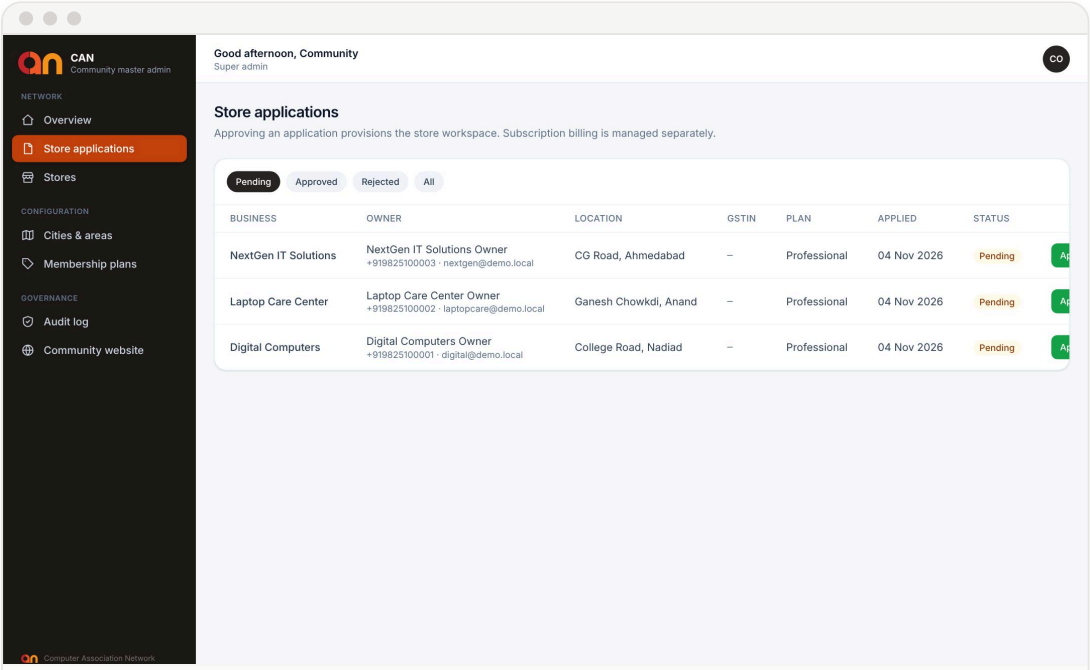
5 · MASTER ADMIN

Running the network

The community team's control room. It shows network-level numbers only — never a store's private customers or invoices.



Network dashboard: stores, approvals, revenue, stores by city, activity



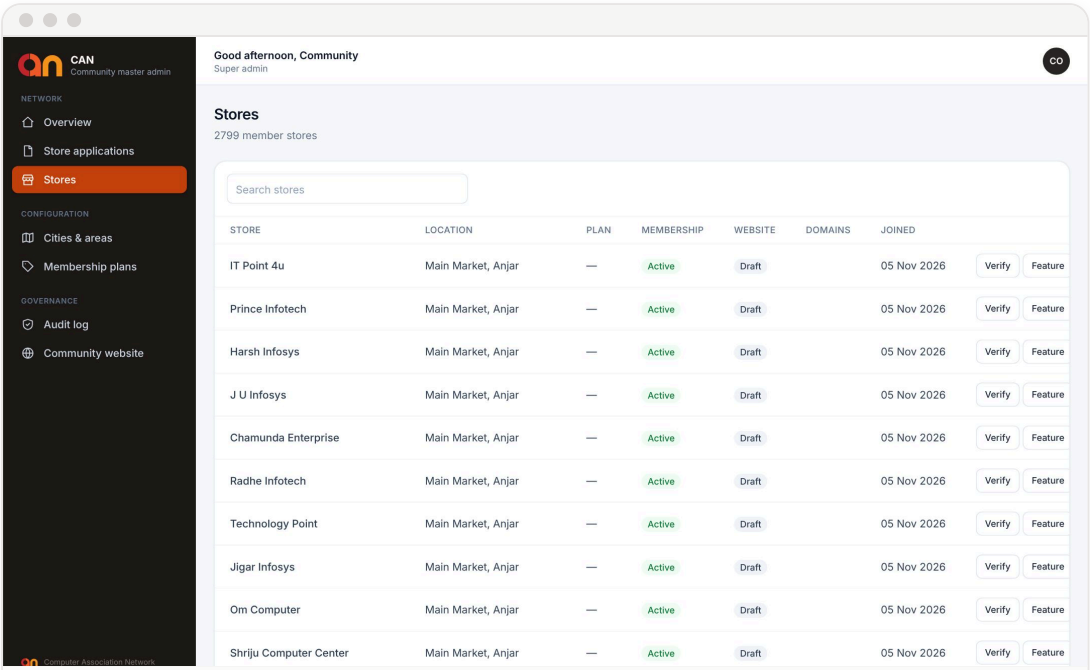
Store applications with approve / reject

- Approve or reject applications; approval provisions the store workspace.
- City and area managers only see and act on their own city/area.
- Every administrative action is written to an append-only audit log.

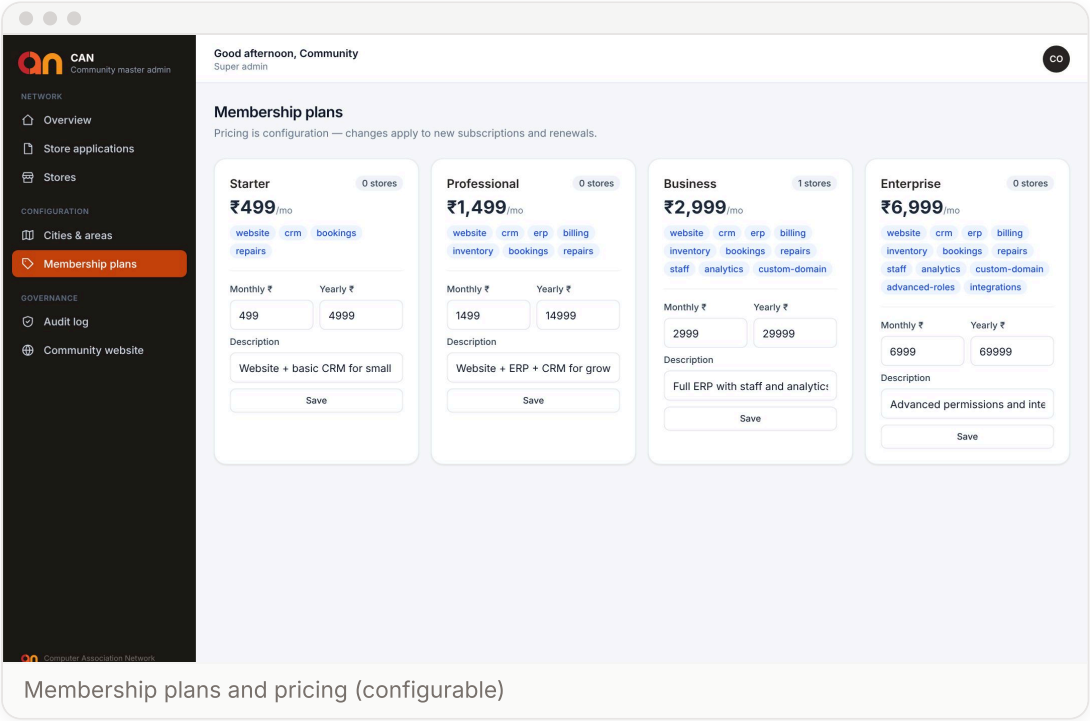
5 · MASTER ADMIN

Stores, cities and plans

Network configuration without developer help.



Stores: plan, status, website, domains; verify, feature, suspend



Membership plans and pricing (configurable)

- Suspending a store immediately closes its workspace and website.
- Cities and areas with GST state codes; plans Starter, Professional, Business, Enterprise with editable prices.

6 · DEMO SCRIPT

A 15-minute demo, step by step

This follows the first-milestone journey from the specification. Use two browser windows: one signed in as admin, one for the shop owner and customer.

#	Who	What to do	What to point out
1	Visitor	Open the home page, scroll associations and leaders, open Computer Association of Nadiad.	The whole Gujarat network in one place.
2	Admin	Cities & areas → add an area to Nadiad.	No developer needed to grow the network.
3	Shop owner	Become a member → fill details, pick CAN as association, choose a plan.	GSTIN and mobile validated.
4	Admin	Store applications → Approve.	Workspace, roles and trial created instantly.
5	Shop owner	Website & branding → colours, hours → Publish. Add a product and a service.	Own website live in minutes.
6	Customer	Open the store website → Book a service → pick a slot.	Real availability; slot can't be double-booked.
7	Shop owner	Bookings → Confirm → Create job card.	Job card number per financial year.
8	Shop owner	Job → Diagnosing → Send quotation.	Customer is notified.
9	Customer	Open tracking link → Approve repair.	No phone call needed.
10	Shop owner	Issue a part from stock → Quality check → Ready.	Stock drops automatically.
11	Shop owner	Create invoice → take payment → Deliver.	GST split, delivery blocked until paid.
12	Shop owner	Dashboard and reports.	Sales, profit and GST summary update.

Security, data and deployment

Store data stays separate

Every store-owned record carries its store ID, and the API checks the user belongs to that store. As a second layer, the PostgreSQL database itself blocks any query from reading or writing another store's rows (row-level security on 23 tables). Automated tests try to break this and must fail.

Money and stock you can trust

GST is calculated in paise with tested rounding. Payments and stock movements are append-only — the database refuses edits; corrections are reversal entries. Invoice and job-card numbers never repeat within a financial year.

Accounts and access

Passwords hashed with Argon2id; sessions rotate and can be revoked; rate limits on login; CSRF protection; role-based permissions per store; city/area scopes for community staff; audit log of admin actions.

Tested

36 automated tests, including the full milestone journey, concurrent double-booking, GST maths, serial sales, payments, CSRF and every cross-store access attempt.

Technology

Next.js + TypeScript + Tailwind (all four interfaces), NestJS API, PostgreSQL with Prisma, Docker. One shared platform engine — new shops are onboarded by configuration, not new code.

Deployment

Runs in Docker next to other apps on the existing VPS, behind the current reverse proxy, with automatic database migrations, a non-superuser database role and a daily backup script.

Data sources and privacy

Associations, office bearers, team, events, gallery, service centres and links come from fitag.in's public pages. Member businesses are listed by name only — their personal mobile numbers, emails and GST numbers are not copied. Leaders' contacts are shown as the association publishes them for their roles. Before public launch, confirm consent with FITAG and the associations.

8 · HONEST STATUS

What is not built yet

Version 1 proves the complete architecture end to end. These parts of the specification are designed for but not yet implemented:

Area	Current state
WhatsApp / SMS / email sending	Messages are queued correctly; the sender only logs them until a provider (WhatsApp Cloud API, SMS gateway, SMTP) is connected.
Online payments & subscription billing	Plans and trial periods exist; Razorpay checkout, renewals and webhooks are not connected.
Customer accounts	Customers book and track via private links; login, “My orders / invoices” portal not built.
Claiming a listing	Owners apply as new members; automatically merging into their existing name-only listing is manual.
Credit notes, returns, expenses	Document types exist; workflows, expenses and double-entry accounting are not built.
HR	Employees and roles exist; attendance, shifts, leave and commissions are not built.
Warranty claims / RMA	Warranty dates are tracked per serial; claim workflow not built.
File uploads	Logos and images are added as links; direct upload storage is not connected.
Admin security	Two-factor login (MFA) for administrators is not yet enabled.
Tax compliance	GST arithmetic is tested, but rates, HSN codes and invoice formats must be validated by a tax professional. E-invoicing is not integrated.

Roadmap

Sizes are relative effort for the suggested team (S ≈ 1–2 weeks, M ≈ 3–5 weeks, L ≈ 6+ weeks). Order follows what is needed to launch, then what makes members stay, then what only a network can offer.

Phase 1 Launch readiness *next 6–8 weeks*

Feature	What it gives members	Size
WhatsApp & SMS	Booking confirmations, repair status, quotation and invoice messages actually delivered (WhatsApp Cloud API + SMS).	M
Razorpay billing	Members pay plan fees online; renewals, reminders, automatic expiry and invoices for the community.	M
Claim your listing	An owner proves ownership (OTP to registered mobile) and converts their name-only listing into a full store.	S
Uploads	Upload logos, product photos and repair intake photos directly (S3 storage, virus scan, image resizing).	S
Customer portal	Customers log in by OTP to see bookings, repairs, invoices and warranties across visits.	M
Admin MFA & backups drill	Two-factor login for admins; tested restore from backup.	S

Phase 2 Complete shop operations *months 3–5*

Feature	What it gives members	Size
Returns & credit notes	Sales returns, credit/debit notes with stock and GST reversal.	M
Expenses & accounts	Expense categories, cash/bank book, receivables/payables, P&L; export to Tally.	L
E-invoice & e-way bill	IRN/QR generation and e-way bills via a GST Suvidha Provider for eligible members.	M
Warranty & RMA	Claims to manufacturer/distributor with status tracking and customer updates.	M
Staff & HR	Attendance, shifts, leave, sales and service commission, technician productivity.	M
AMC contracts	Annual maintenance contracts with visit schedules, reminders and renewals.	M
Offers & reviews	Store promotions on websites and directory; verified customer reviews after delivery.	S

Phase 3 Network-only features *months 6–9*

Feature	What it gives members	Size
Member-to-member trade	Dead-stock exchange and B2B marketplace between members (fitag.in already has “Dead Stock”).	L
Distributor price lists	Authorised distributors publish price lists and schemes to all members; purchase orders in one click.	L
Association management	Membership dues, circulars, meeting and event registration, voting/elections per association.	M
Fraud & credit alerts	“Take care” alerts shared across associations about fraud cases and defaulters (moderated).	S
Network analytics	For FITAG: market trends by city, category and brand (anonymised, opt-in).	M
Gujarati & Hindi	Full Gujarati and Hindi interface for owners, staff and customers.	M
Mobile apps	Android/iOS apps for owners (sales, repairs, approvals) and technicians (job updates, photos).	L
AI assistant	Answers “what sold best this month?”, drafts quotations and WhatsApp replies, suggests reorders.	M

Next steps

1. Review this demo with the CAN / FITAG committee and collect feedback.
2. Confirm consent for using the association member lists and leaders' published details.
3. Connect a real domain (e.g. can.org.in) and WhatsApp/SMS provider.
4. Pilot with 10–20 Nadiad members; use their feedback to prioritise Phase 1.
5. Onboard the remaining associations zone by zone.

One network. Independent businesses. One powerful ecosystem.